Practice Lab:

Edit and Compile using Eclipse with Scala

- This Practice Lab has as a prerequisite, Practice Lab 6241, where we install and configure Eclipse Photon on MacOS including a Scala plugin.
- This Practice Lab also has a prerequisite, Discussion Units 7544/7545 and 7548/7549, where we make the DSE Analytics client we run here.

- In this Practice Lab, we edit, compile, and run a DSE Analytics client program. As such, this lab expects an operating DSE cluster with DSE Analytics be accessible.

- Instructions in this practice lab are written for MacOS, but should function for most Linux distros.

- This Practice Lab will require reasonable Internet access.

Eclipse IDE: What is it ?

Open source integrated developer's workbench (IDE)

- From 1984's Visual Age, a similar IBM program
- Not apache, is Eclipse Public License (ECL)
- Releases with science related names
- June/2018 release is Photon
- Many/many development languages supported
- Extensible (Vi plugin !)

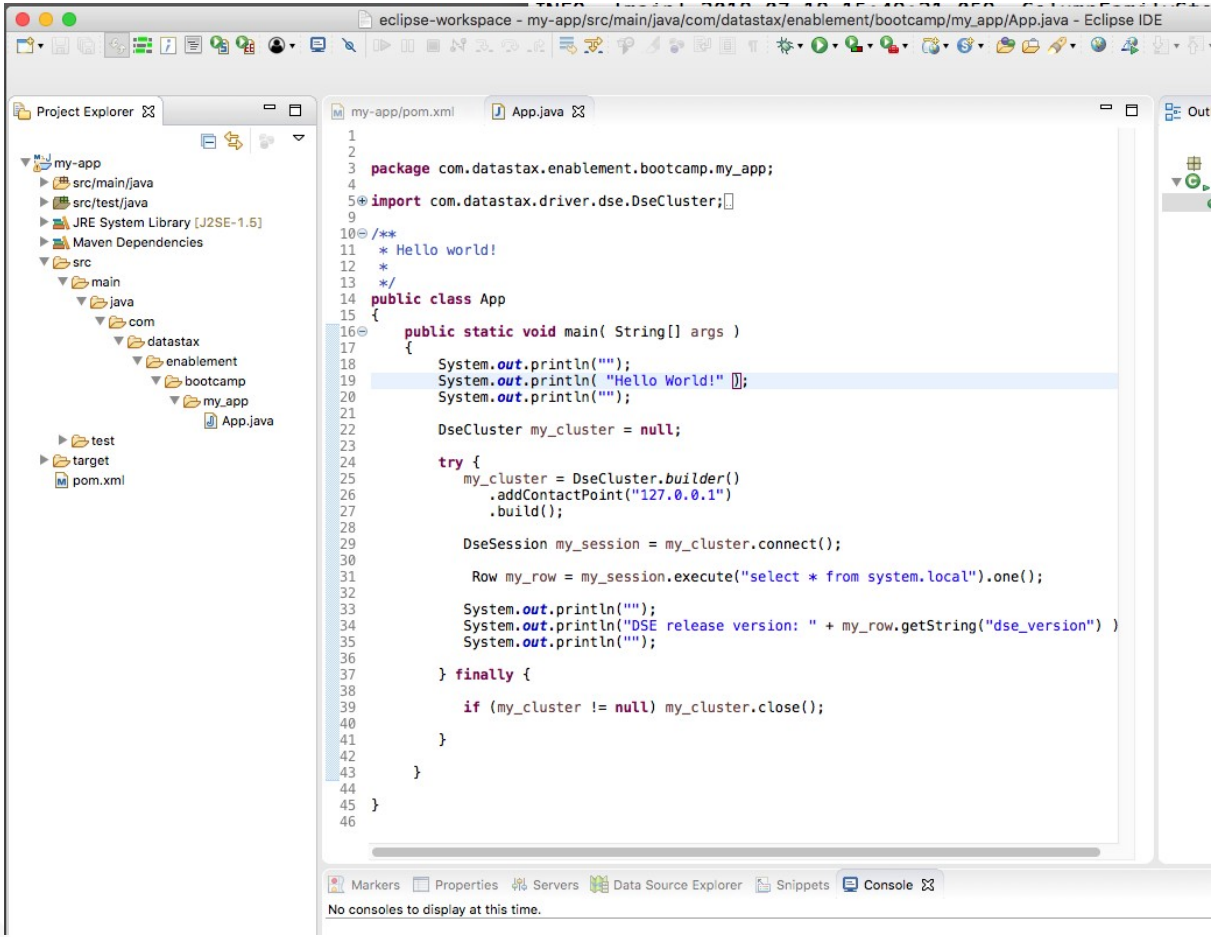
Eclipse: How to Install

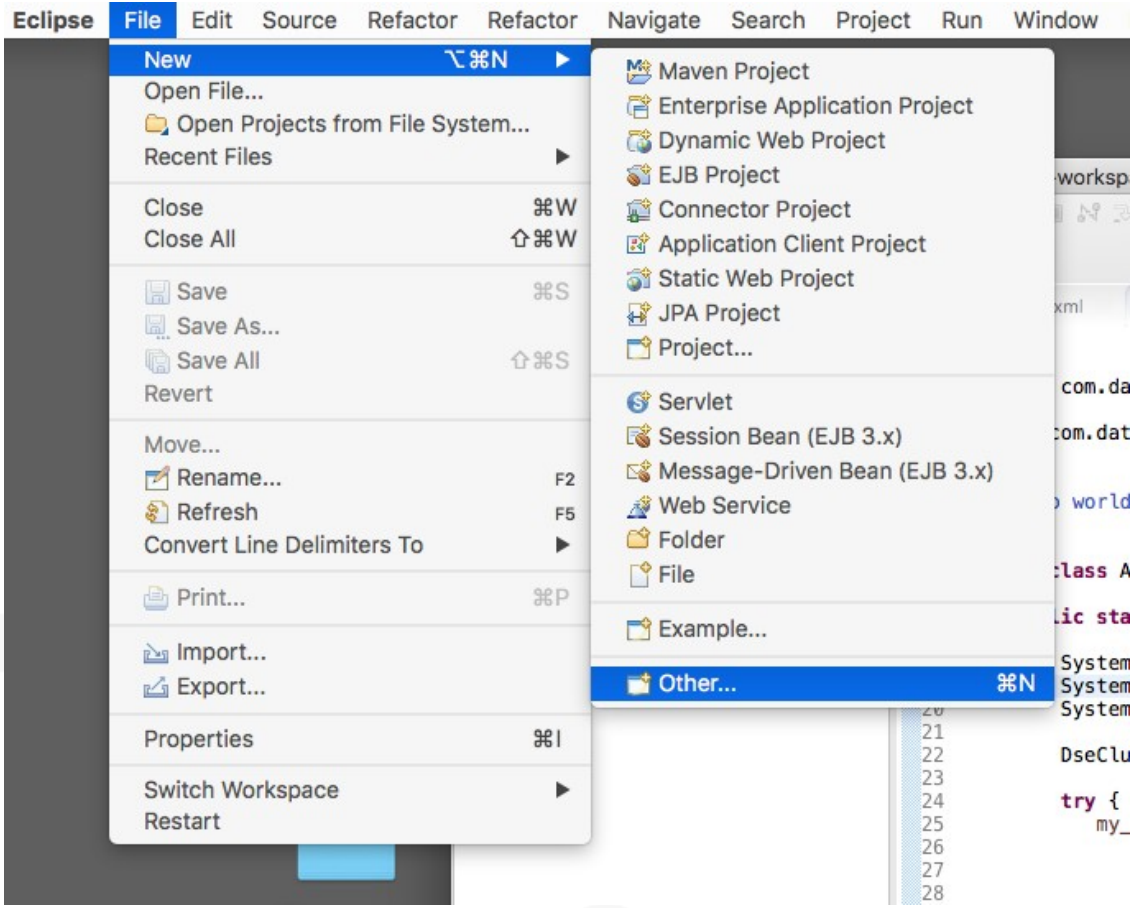
Instructions in Practice Lab 6241.

- Reference Practice Lab 6241, where we installed Eclipse, and an Eclipse plugin specific to Scala.
- Launch Eclipse, and then ..

After Practice Lab 6241-

- Configured for DSE Java clients
- Eclipse Scala plugin installed





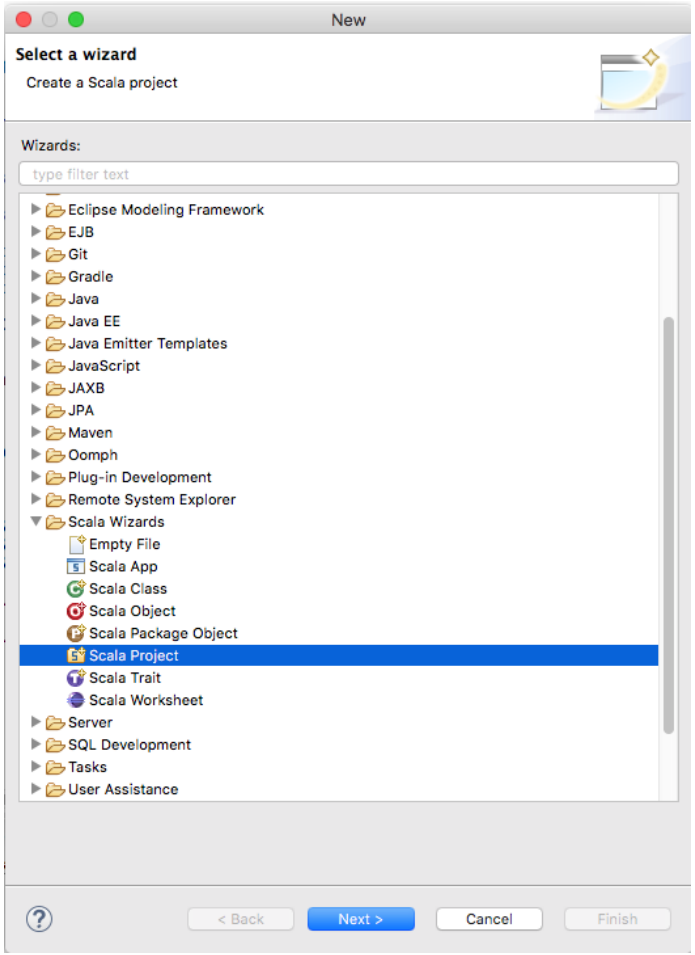
From the Eclipse Menu Bar-

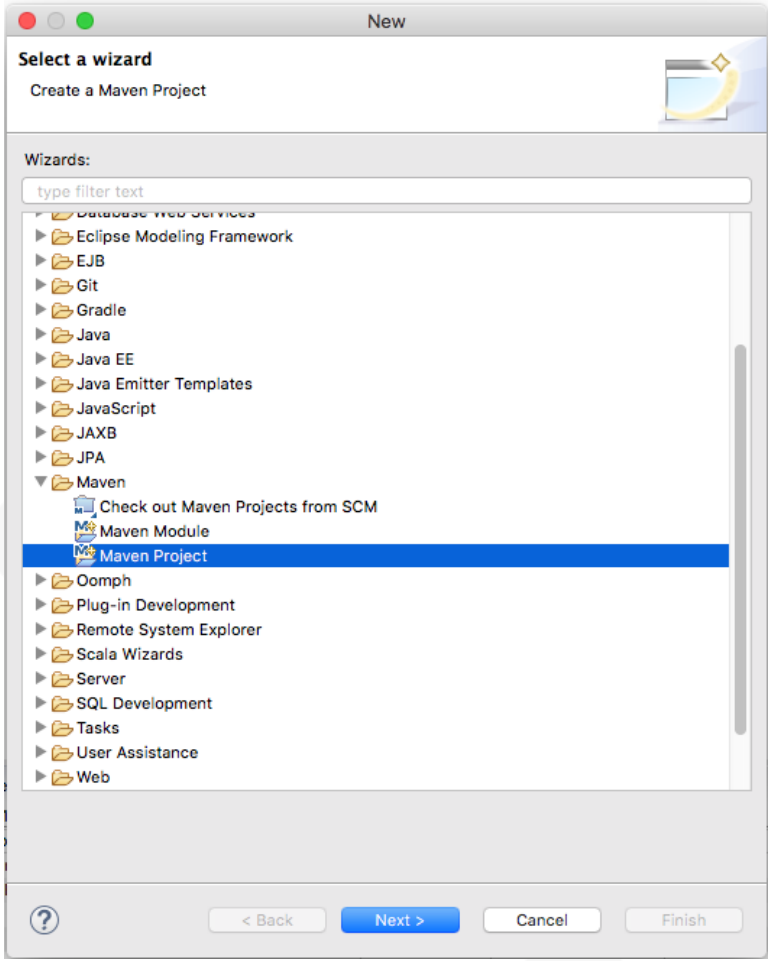
- File -> New -> Other

Don't use this-

- Simple Scala programs only
- Not a Maven project, no pom.xml

- Better off using the DSE Spark REPL



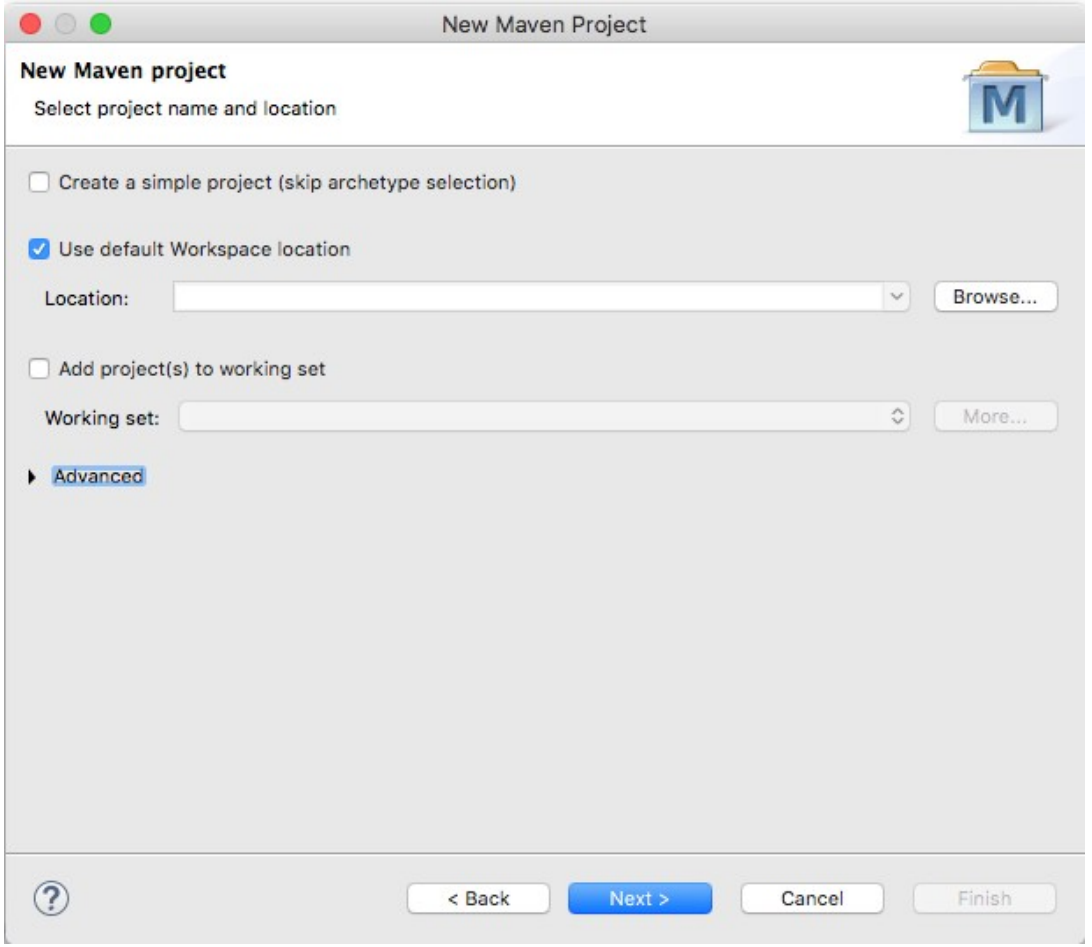


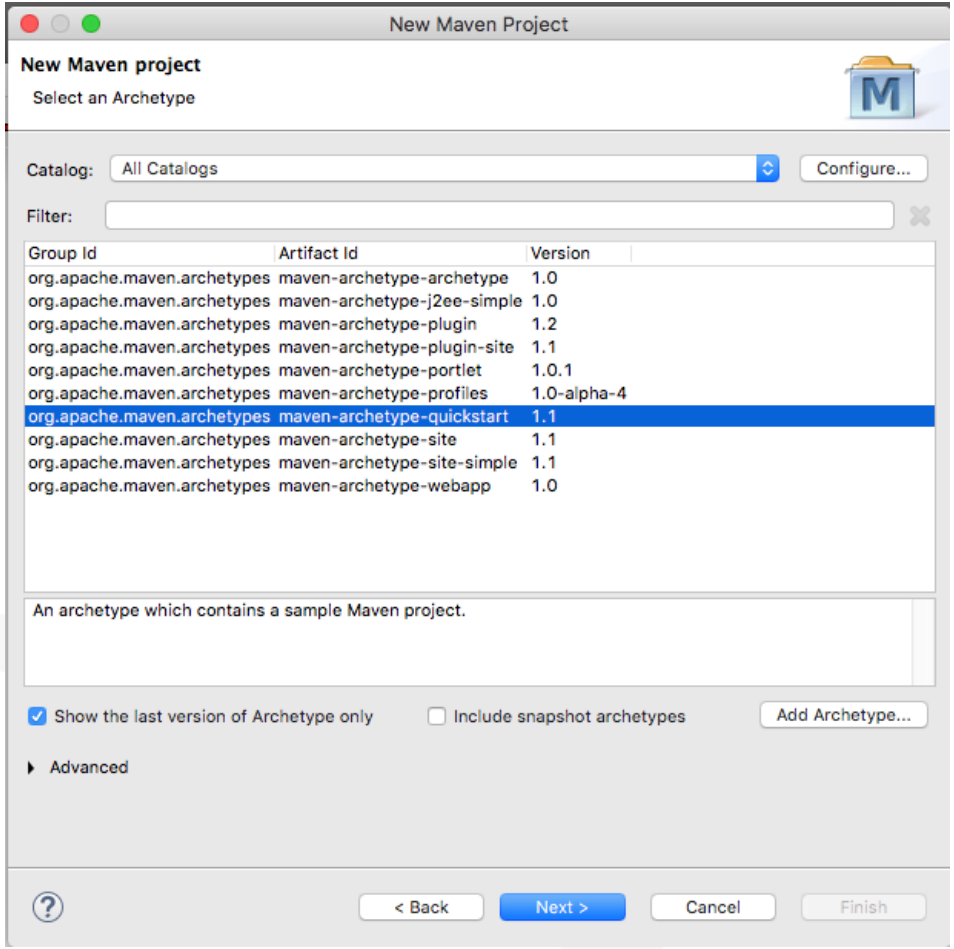
In the, Select a wizard, modal dialog box

- Maven -> Maven Project
- This option gives us support for a pom.xml, which will automatically pull all of our (DSE and Spark) dependencies

Just click through this dialog box-

- There's nothing we need on this dialog box-
- Click, Next



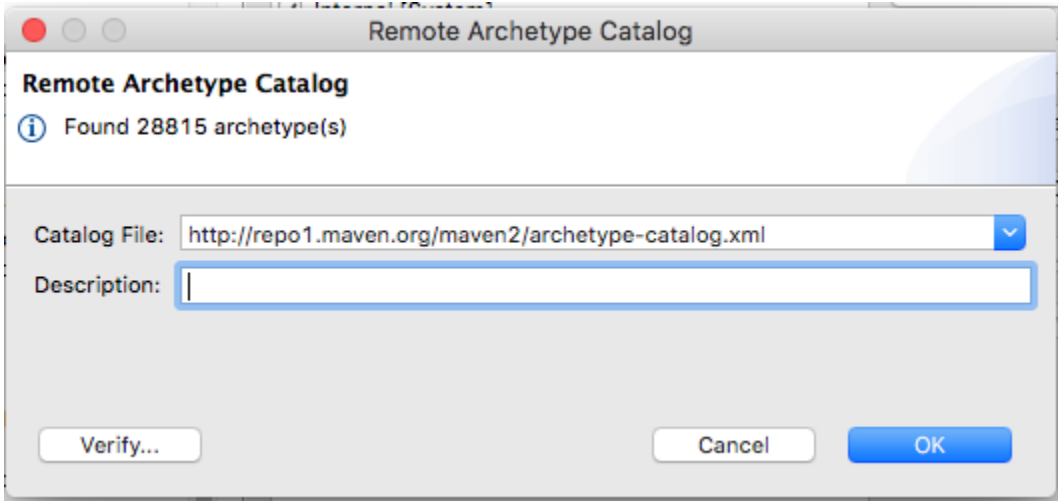


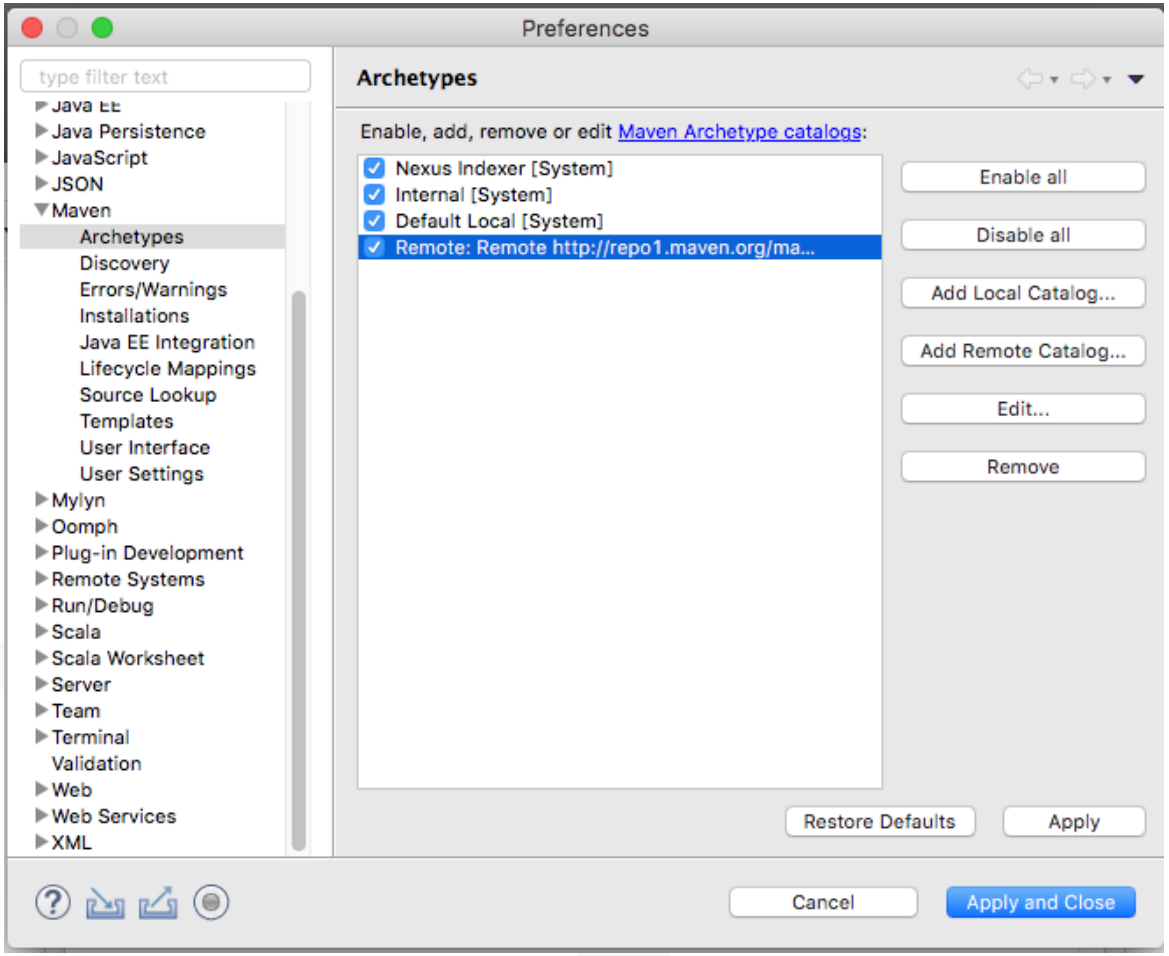
Maven Archetypes

- Maven Archetypes, a project templating device; filesystem structure, other
- The default is the same we used in Discussion Unit 6240, the default Java client archetype
- We want Scala
- Click, Configure

Online (Remote) Maven Archetype Repository

- Enter the value as shown in the text entry field titled, Catalog File
- Optionally Click, Verify
- And Click, OK



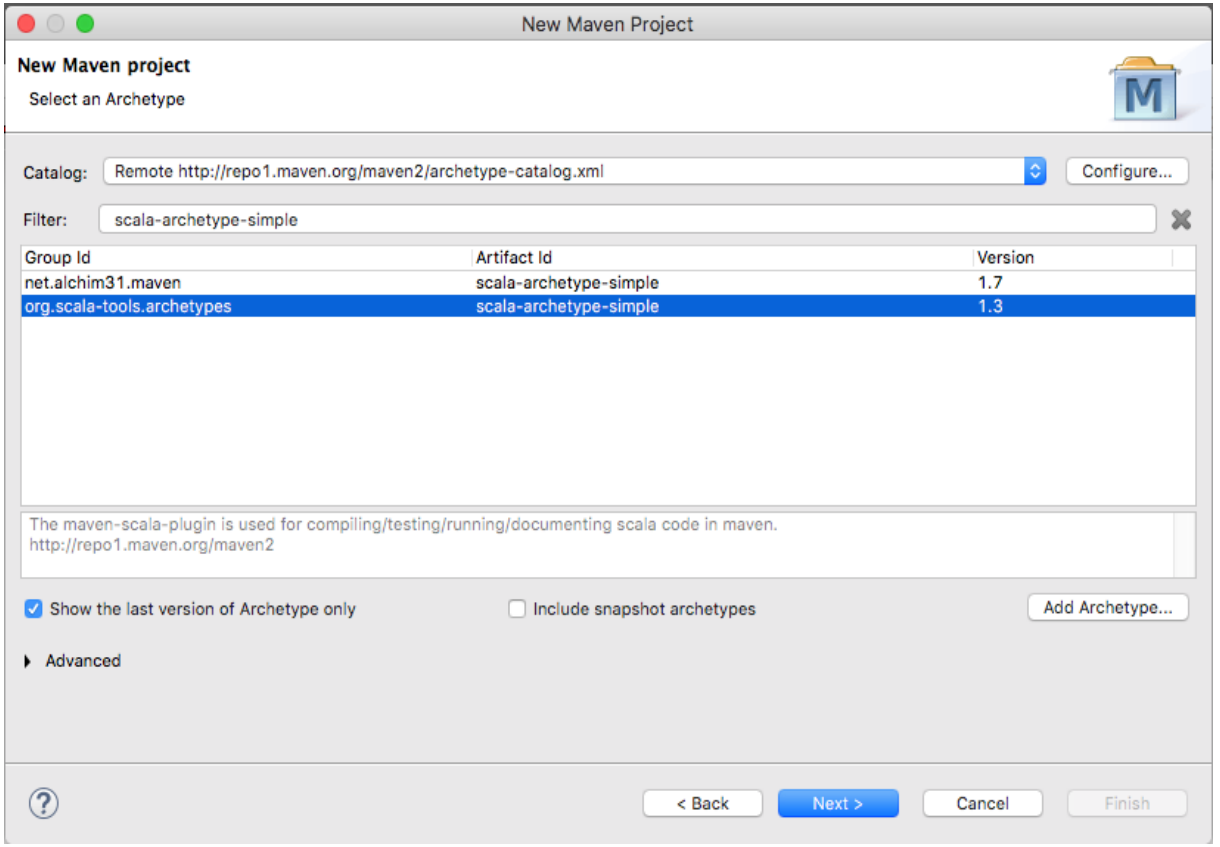


Archetype Catalog added-

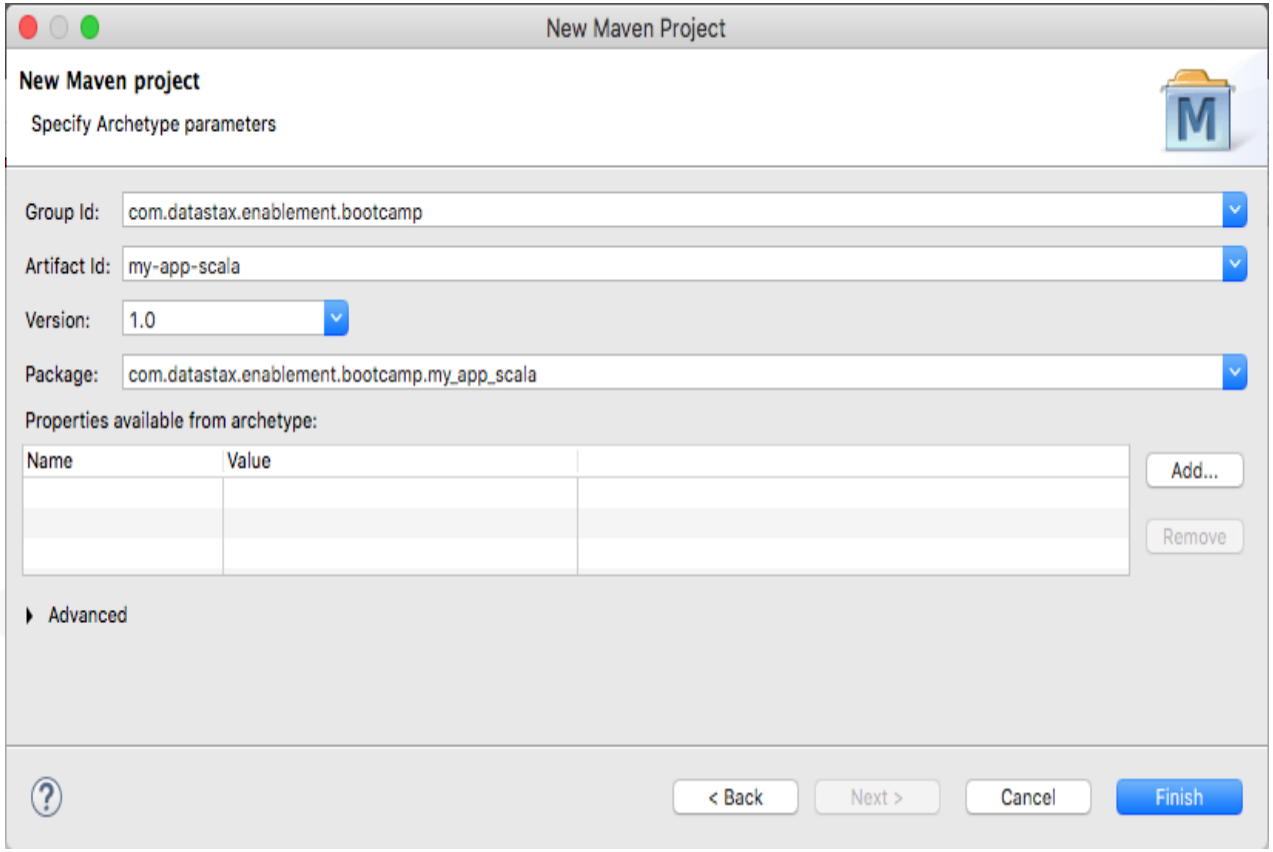
- After the addition of our Archetype Catalog, Click, Apply and Close

Picking a given Archetype

- Manage the display as shown on the right-
- Specifically enter a Filter of, scala-archetype-simple, to reduce the number of options
- When your dialog box matches that as shown, Click, Next



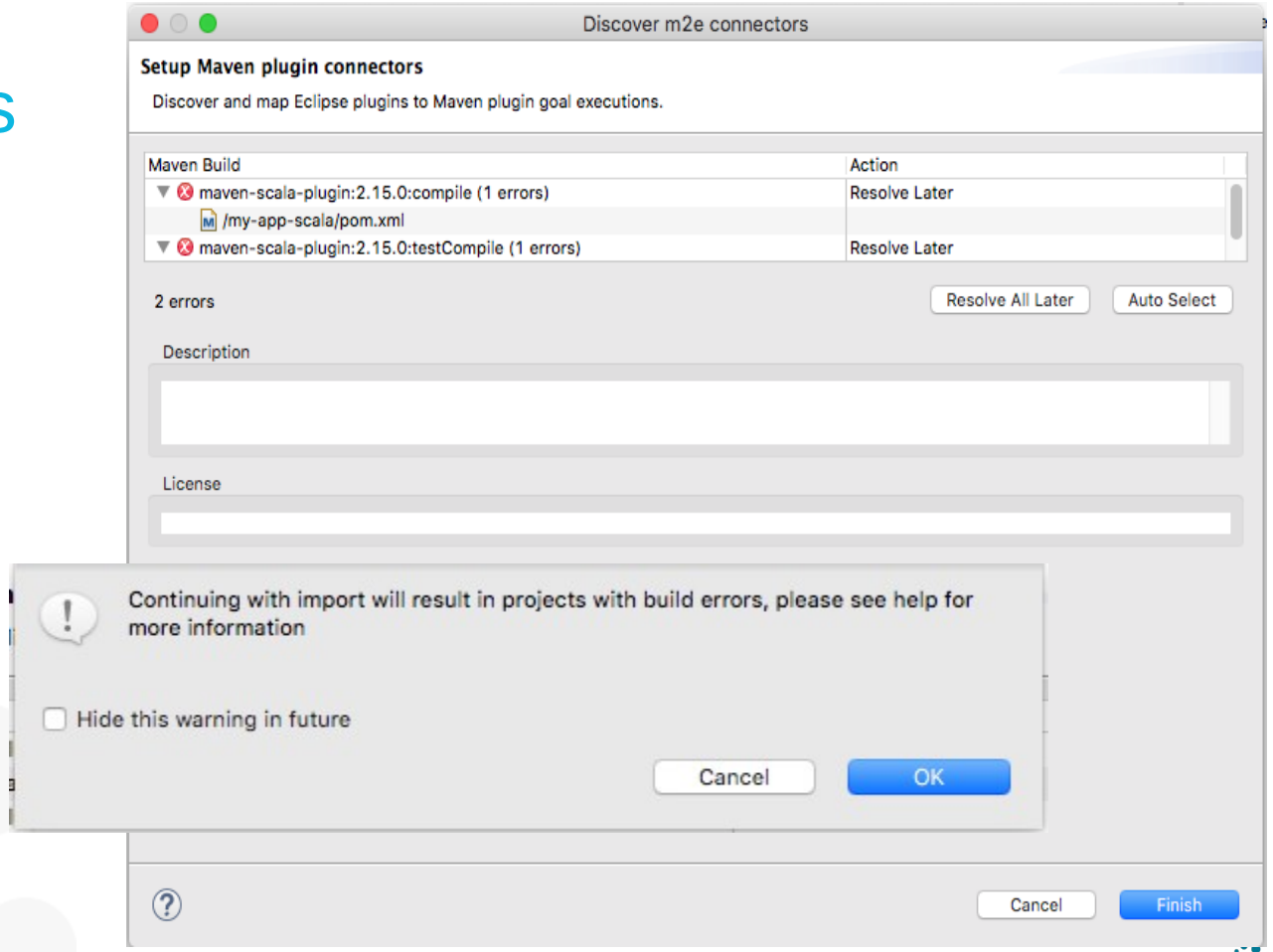
The Project Proper-

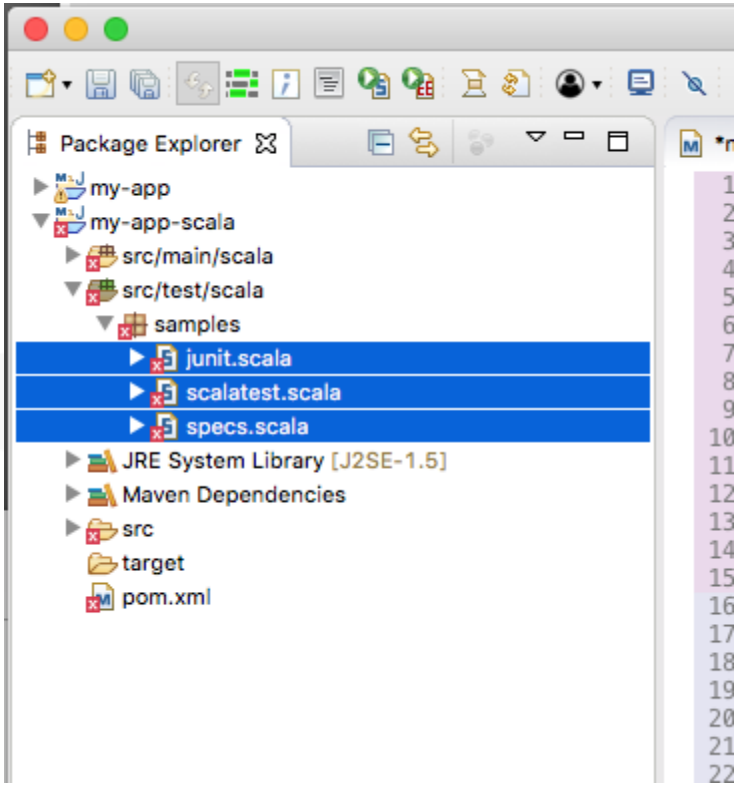


- Enter a GroupId, ArtifactId, and Version as shown
- Click, Finish

As generated, a number of things are broken-

- We'll fix these; not here

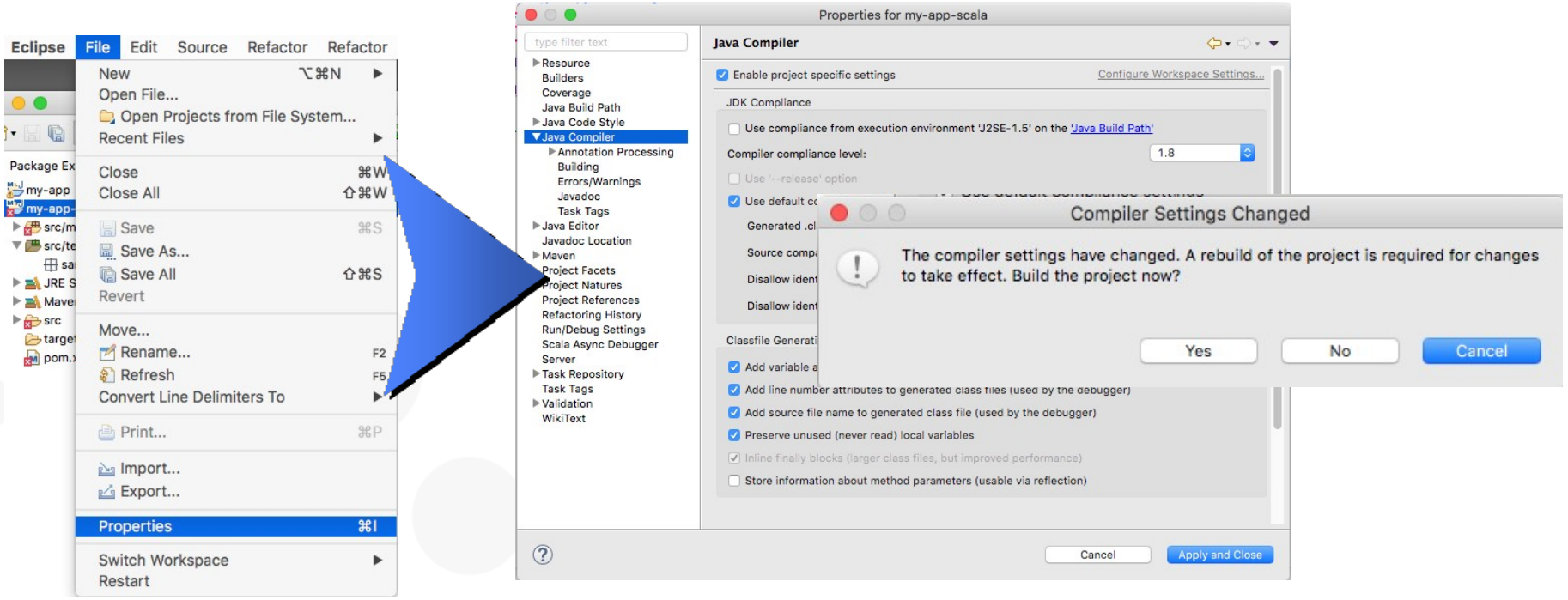




Delete the generated/stub, test code

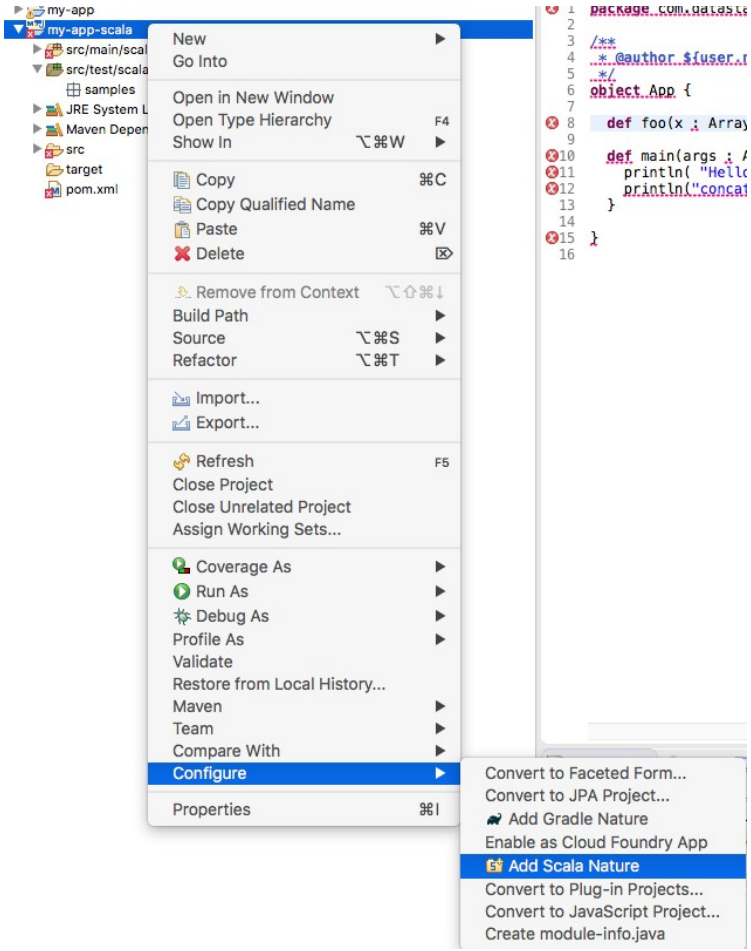
- In the Project Explorer view, navigate to, my-app/scala -> src/test/scala -> samples
- Shift-Click all entries, Right-Click, and Delete
- These are stub test units and do not compile

Setting Java Compiler version to 1.8-

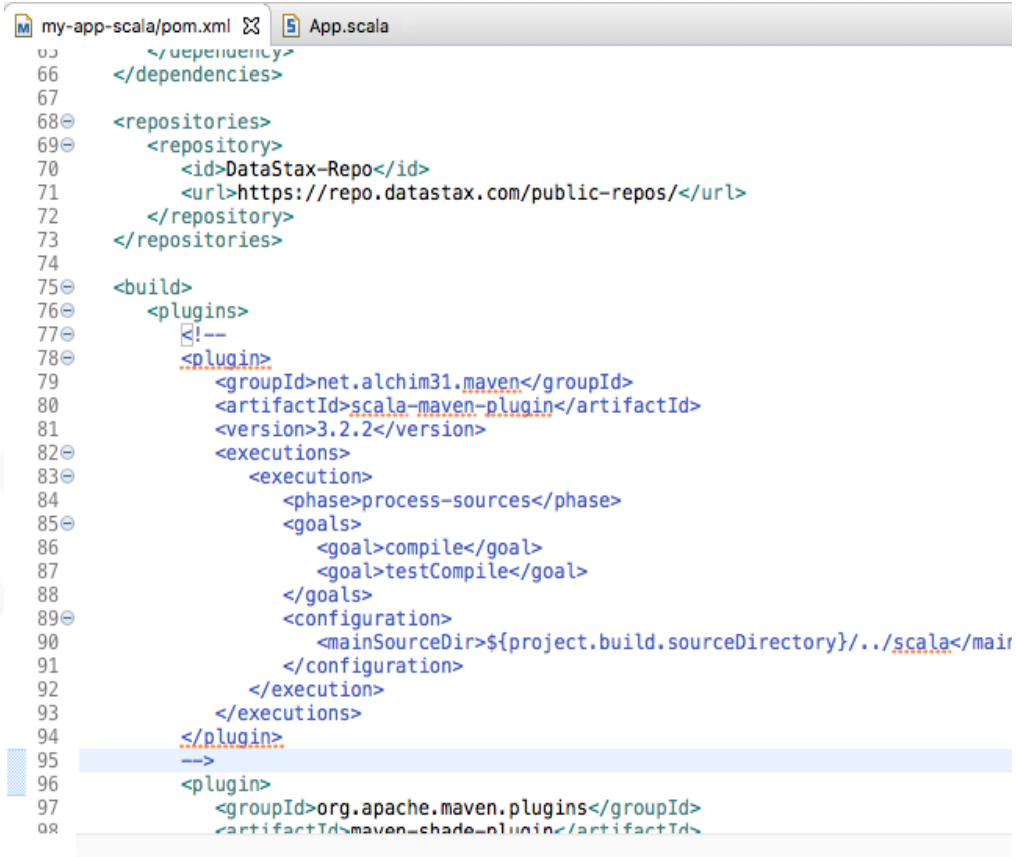


Maven knows, Eclipse doesn't know

- While Maven knows we want a Scala project, Eclipse does not
- In the Project Explorer view, Right-Click the project, and select; Configure -> Add Scala Nature



Fix the pom.xml-



- Edit the pom.xml file as detailed on the Notes page.
- Save
- From the Menu Bar; Project -> Clean
- (Clean can take a bit of time.)

What is a pom.xml ?

- xml
- "project object model"
- Each project has one

- Tells Maven how to compile, etc, your project

- Edit once and done ?

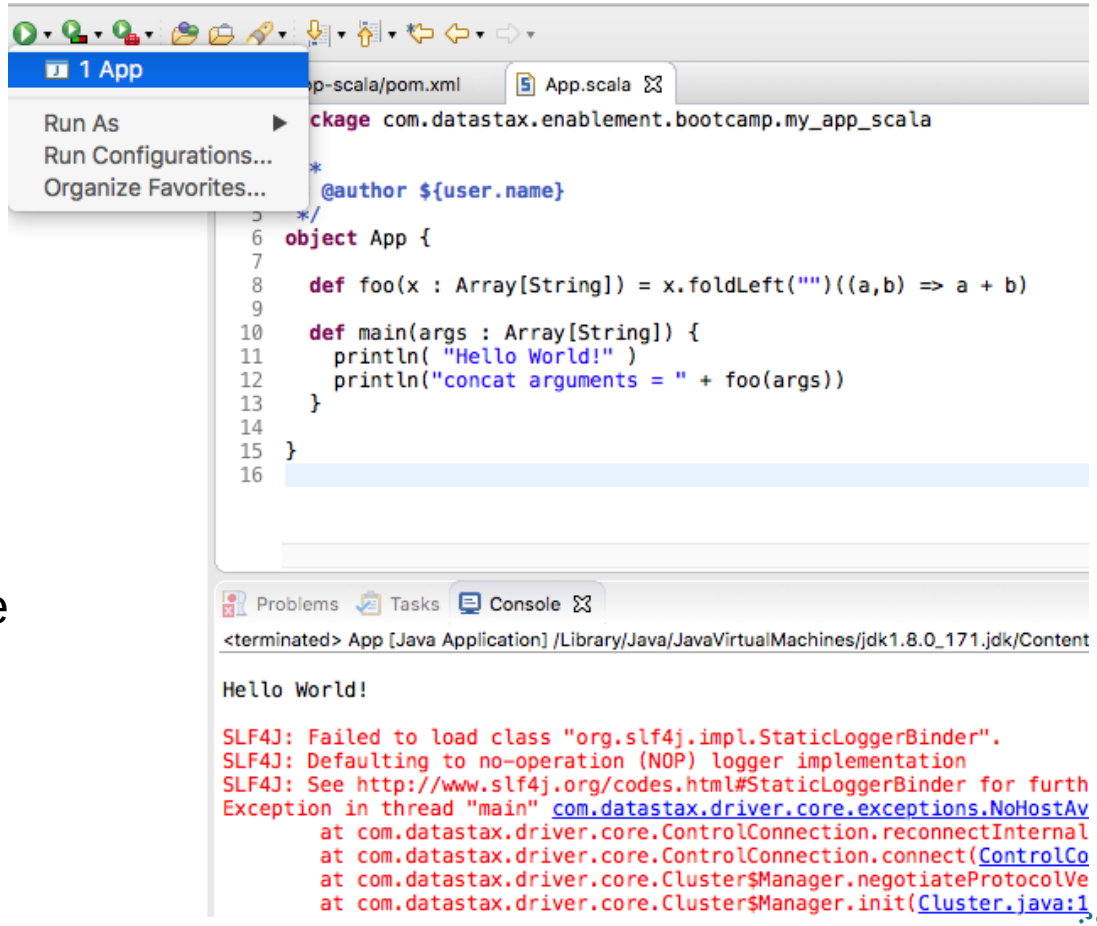
```
<project xmlns="http://maven.apache.or ...
  xsi:schemaLocation="http://mav ...
  <modelVersion>4.0.0</modelVersion>

<groupId>com.datastax.enablement.bootcamp<
/groupId>
  <artifactId>my-app</artifactId>
  <packaging>jar</packaging>
  <version>1.0-SNAPSHOT</version>
  <name>my-app</name>
  <url>http://maven.apache.org</url>
  <dependencies>
    <dependency>
      <groupId>junit</groupId>
      <artifactId>junit</artifactId>
      <version>3.8.1</version>
      <scope>test</scope>
    </dependency>
  </dependencies>
</project>
```

Run the generated App.scala

- Time for a test-
- In the Project Explorer view, find and open (Double-Click) App.scala

- From the Eclipse Toolbar, click the Run icon (Run is context sensitive, and App.scala must be current.)

- Successful output as shown in the Console view



That was the default generated App.scala-

```
: All host(s) tried for query failed (tried: /127.0.0.1:9042 (com.datastax.driver.core.exceptions
on.java:237)
```

- From Discussion Unit 7548/7549, and 7554, grab the very same pom.xml, and App.scala you created and ran there.
- Copy and paste the above files into the same pom.xml and App.scala you just generated and ran. Be certain to Save.
- Do a, Project -> Clean, and Run

```
99        // ------------------------------
100
101
102       val My_Schema = StructType(Array(
103          StructField("customer_num"    , IntegerType, true),
104          StructField("fname"           , StringType, true),
105          StructField("lname"           , StringType, true),
106          StructField("company"         , StringType, true),
107          StructField("address1"        , StringType, true),
108          StructField("address2"        , StringType, true),
109          StructField("city"            , StringType, true),
110          StructField("state"           , StringType, true),
111          StructField("zipcode"         , StringType, true),
112          StructField("phone"           , StringType, true)
```

Problems Tasks Console

```
<terminated> App [Java Application] /Library/Java/JavaVirtualMachines/jdk1.8.0_171.jdk/Contents/Home/bin/java (Jul 18, 20

Hello World!

SLF4J: Failed to load class "org.slf4j.impl.StaticLoggerBinder".
SLF4J: Defaulting to no-operation (NOP) logger implementation
SLF4J: See http://www.slf4j.org/codes.html#StaticLoggerBinder for further details.

DSE release version: 6.0.0
```

You are done when-

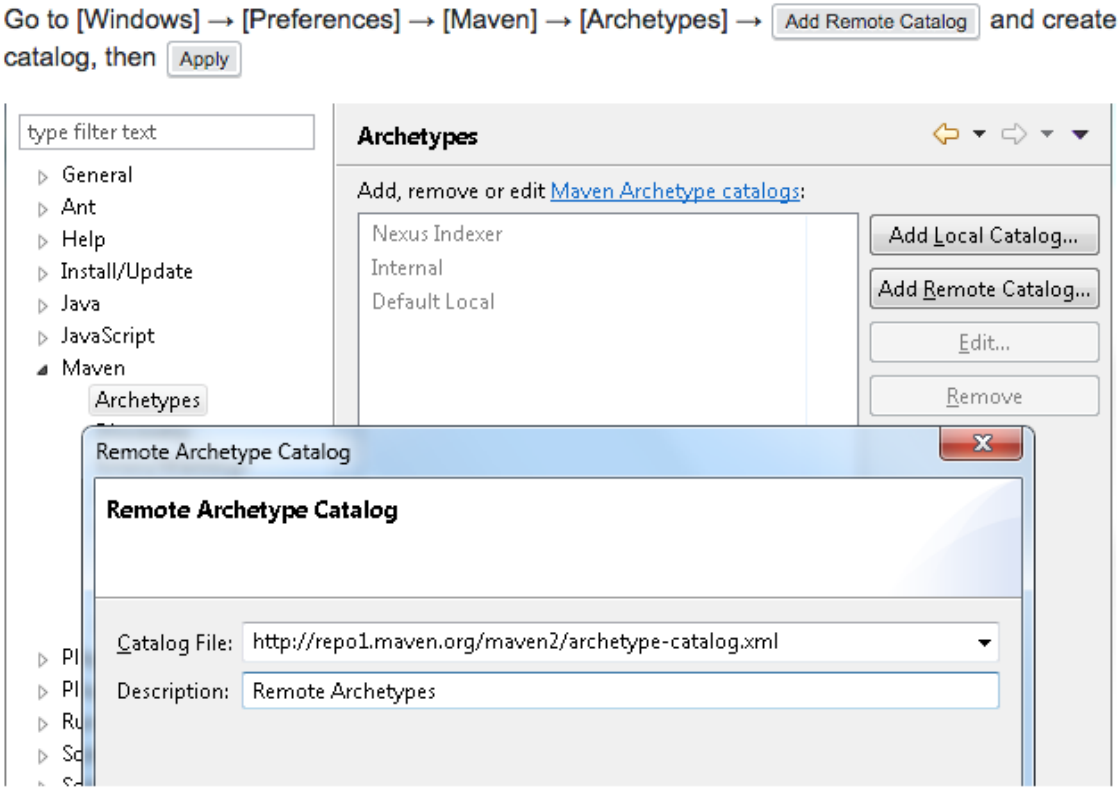
Practice Lab:

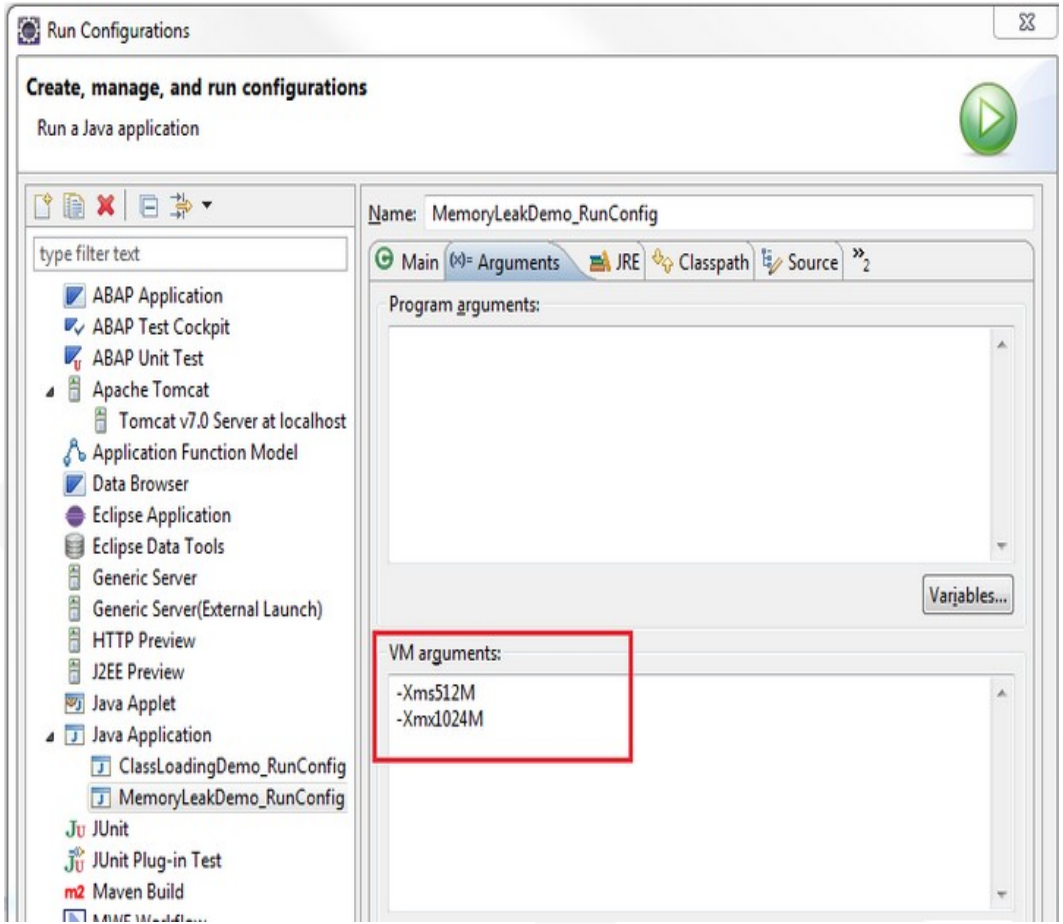
Lessons Learned

Additional Detail:

More Help Finding Archetypes:

The Notes page lists a Url to a StackOverflow article on finding Maven Scala archetypes-





Eclipse Sluggish ?

- Eclipse is a Java app
- Tune the JVM settings as shown; specifically Xmx
- 1G default is common, 4G if you have room
- Url on Notes page